

"It has today occurred to me that an amplifier using semiconductors, rather than vacuum, is in principle possible."

—William Shockley's Laboratory notebook, as written on 29 Dec. 1939

Overclocking helps to improve value for money

Those who are familiar with overclocking can save a good amount of money while having some fun. In overclocking, you increase the operating frequency of your CPU and memory in order to get a faster computer. It involves tweaking the core voltage and cooling systems as well so that your faster machine remains reliable. A bad overclock can cause your computer to overheat and shut down, among other things.

AMD Ryzen Master Utility can be used to overclock easily on Ryzen chips. For example, Ryzen 7 1700 can be worked on and clocked up to compete with the much more expensive 1800X! With Intel silicon, an alternative is to get Core i7-7700K Kaby Lake-S processor. This chip works with a cheaper Z270 motherboard and lower TDP of 91W, and can be overclocked to almost 5GHz if you have the skills and patience.

If you are looking for even more value for money, AMD Ryzen 5 1600 is a better solution than similarly priced Intel Core-i5 7500 chips since AMD chips come with 12 threads as against Intel chips' just four threads. This should not be a problem for older software that rely on single-thread performance, but will significantly impact others.

Do you need heavyweight processing?

The new Ryzen Threadripper 1950X is a 16-core, 32-thread monster with 64 PCIe lanes that run its cores at 4GHz on Turbo. While the maximum temperature is 68°C, its default TDP is a whopping 180W!

Ryzen chips are also said to inte-

grate some machine learning, which allows the CPU to perform better in recurring tasks. On the other hand, Intel's 10-core Core-i9 7900X runs at 4.3GHz with TDP of 140W. It comes with 44 PCIe 3.0 lanes and supports AVX-512.

Server-grade capabilities available here

A scalable foundation for the modern data centres requires interconnecting on-chip components to improve efficiency and scalability of multi-core processors. Intel's new mesh network allows scalability within one large piece of silicon, while AMD's multi-silicon strategy uses its new Infinity Fabric for scale.

Both these can be seen as evolutions of the Front Side Bus (FSB) or HyperTransport technology. The Infinity Fabric allows full utilisation of the DRAM on a system-on-chip (SoC) or GPU, which means theoretical performance limits will be achievable easily.

AMD's EPYC and Ryzen are built with blocks called CPU Complex (CCX), which consist of 'Zen' cores connected with an L3 cache. Inter-

estingly, while each core has its own 2MB of L3 cache, it can also access the other 6MB of cache quickly.

There are two CCX blocks in each die, while one complete EPYC chip is a multi-chip module (MCM) made of four such dies. The EPYC 7601 has 32 cores running at a maximum frequency of 3.2GHz, 64MB of L3 cache and TDP of 180W.

In the SkyLake-SP Xeon Scalable Processor family, Intel added 768kB of L2 cache per core, changed the way L3 cache works and added a second 512-bit AVX-512 unit. Intel AVX-512 enables twice the number of floating-point operations per second (FLOPS) per clock cycle compared to what we had before.

Anandtech notes that "even in the best-case scenario, some of the performance advantage will be negated by the significantly lower clock speeds (base and turbo) that Intel's AVX-512 units run at due to the sheer power demands of pushing so many FLOPS."

The Intel Xeon Platinum 8180 has 28 cores and 56 threads being pushed at 3.80GHz, 380.5MB of L3 cache and TDP of 205W.



AMD Ryzen Threadripper